

Kingdom of Saudi Arabia
Ministry of Education
College of Computer
Computer Science Department

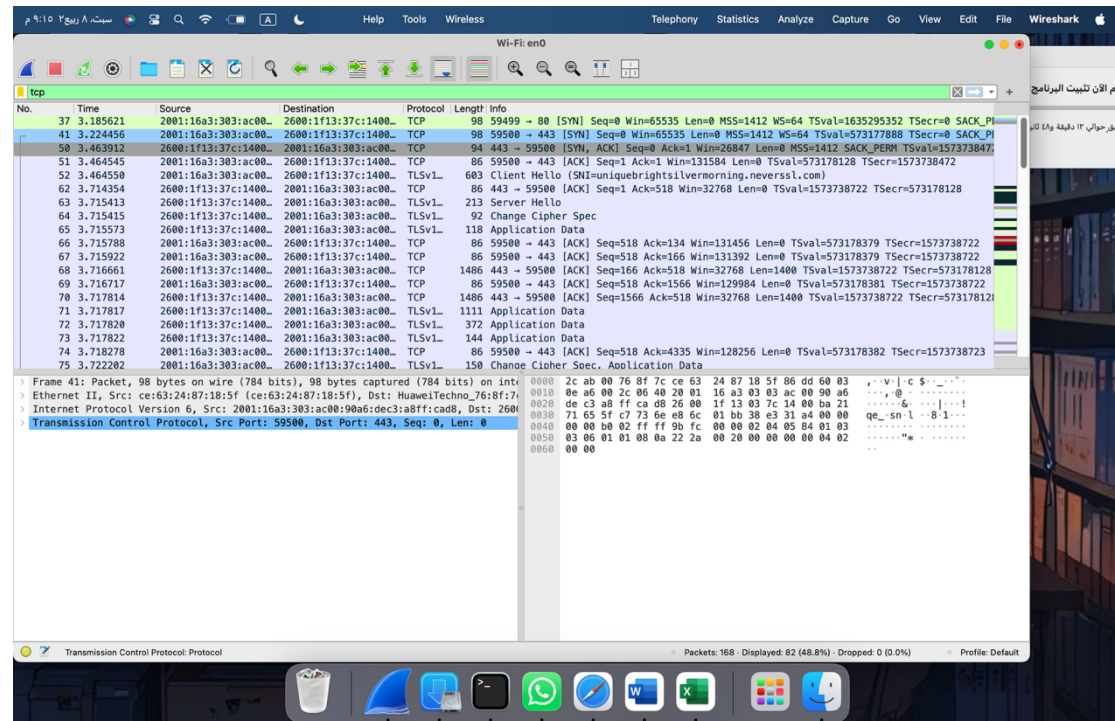


المملكة العربية السعودية
وزارة التعليم
كلية الحاسب
قسم علوم الحاسب

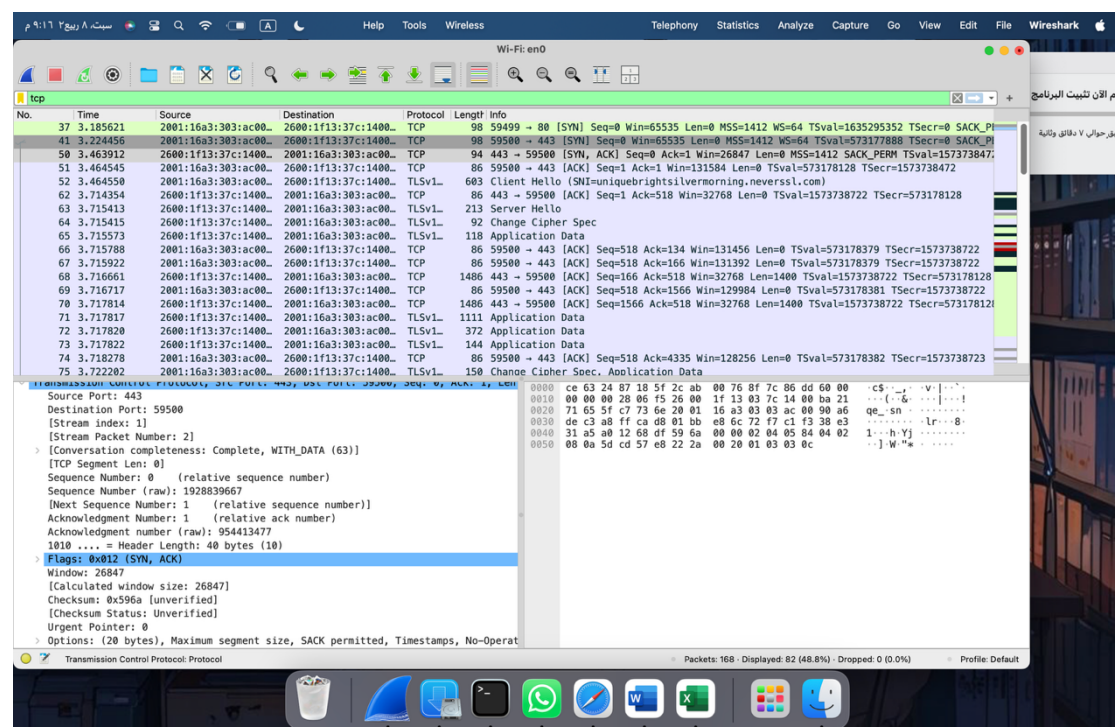
Course: CS471 - Web Technologies
Lab Number: Lab 1 , The Internet Protocols
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Part 1 & 2: Capturing and Analyzing HTTP & TCP Traffic

1. TCP Three-Way Handshake



2. Sequence and Acknowledgment Numbers



3. HTTP Request and Response Messages

The image shows a Wireshark capture of HTTP traffic on the Wi-Fi interface. The packet list on the left shows several HTTP packets. The selected packet is packet 131, which is a GET request for /online HTTP/1.1. The packet details pane on the right shows the structure of the HTTP message, including the request line, headers, and body. The packet bytes pane at the bottom shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
131	11.985920	2001:16a3:303:ac00::	2600:1f13:37c:1400::	HTTP	485	GET /online HTTP/1.1
133	12.204591	2600:1f13:37c:1400::	2001:16a3:303:ac00::	HTTP	671	HTTP/1.1 301 Moved Permanently (text/html)
135	12.211327	2001:16a3:303:ac00::	2600:1f13:37c:1400::	HTTP	486	GET /online/ HTTP/1.1
138	12.585855	2600:1f13:37c:1400::	2001:16a3:303:ac00::	HTTP	232	HTTP/1.1 200 OK (text/html)
142	12.537973	2001:16a3:303:ac00::	2600:1f13:37c:1400::	HTTP	464	GET /favicon.ico HTTP/1.1
143	12.866781	2600:1f13:37c:1400::	2001:16a3:303:ac00::	HTTP	502	HTTP/1.1 200 OK (PNG)

Part 3: Capturing and Analyzing UDP Traffic

The image shows a Wireshark capture of UDP traffic. The packet list on the left shows a series of UDP packets. The selected packet is packet 1, which is a UDP packet from 2001:16a3:303:ac00:: to 2001:4860:4802:321::. The packet details pane on the right shows the structure of the UDP message, including the source and destination ports, length, and checksum. The packet bytes pane at the bottom shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	2001:16a3:303:ac00::	2001:4860:4802:321::	UDP	775	64463 → 443 Len=713
2	0.000263	2001:16a3:303:ac00::	2001:4860:4802:321::	UDP	909	64463 → 443 Len=847
3	0.102473	2001:4860:4802:321::	2001:16a3:303:ac00::	UDP	96	443 → 64463 Len=34
4	0.115631	2001:4860:4802:321::	2001:16a3:303:ac00::	UDP	910	443 → 64463 Len=848
5	0.117919	2001:16a3:303:ac00::	2001:4860:4802:321::	UDP	95	64463 → 443 Len=33
6	0.117927	2001:16a3:303:ac00::	2001:4860:4802:321::	UDP	93	64463 → 443 Len=31
7	0.117928	2001:16a3:303:ac00::	2001:4860:4802:321::	UDP	93	64463 → 443 Len=31
9	0.217136	2001:4860:4802:321::	2001:16a3:303:ac00::	UDP	92	443 → 64463 Len=30
10	1.086455	2001:16a3:303:ac00::	2a02:267f:58:0:ace::	UDP	342	59963 → 443 Len=280
11	1.087567	2001:16a3:303:ac00::	2a02:267f:58:0:ace::	UDP	342	59963 → 443 Len=280
12	1.088540	2001:16a3:303:ac00::	2a02:267f:58:0:ace::	UDP	342	59963 → 443 Len=280
13	1.892655	2001:16a3:303:ac00::	2a01:b740:a41:280::	QUIC	1262	Initial, DCID=ac67835e407e17e, PKN: 0, CRYPTO, PADDING
14	1.164325	2a02:267f:58:0:ace::	2001:16a3:303:ac00::	UDP	93	443 → 59963 Len=31
15	1.174358	2a02:267f:58:0:ace::	2001:16a3:303:ac00::	UDP	463	443 → 59963 Len=401
16	1.174621	2a02:267f:58:0:ace::	2001:16a3:303:ac00::	UDP	640	443 → 59963 Len=578
17	1.174998	2a02:267f:58:0:ace::	2001:16a3:303:ac00::	UDP	358	443 → 59963 Len=296
18	1.175294	2001:16a3:303:ac00::	2a02:267f:58:0:ace::	UDP	185	59963 → 443 Len=43
19	1.191277	2a01:b740:a41:280::	2001:16a3:303:ac00::	QUIC	1262	Handshake, SCID=001ac3eed4bb169670e26a7471fb0f0afde01322
20	1.191766	2a01:b740:a41:280::	2001:16a3:303:ac00::	QUIC	1262	Handshake, SCID=001ac3eed4bb169670e26a7471fb0f0afde01322
21	1.191769	2a01:b740:a41:280::	2001:16a3:303:ac00::	QUIC	1262	Handshake, SCID=001ac3eed4bb169670e26a7471fb0f0afde01322
22	1.192405	2001:16a3:303:ac00::	2a01:b740:a41:280::	QUIC	1262	Initial, DCID=001ac3eed4bb169670e26a7471fb0f0afde01322, PKN: 1, ACK, PADDING
23	1.192413	2001:16a3:303:ac00::	2a01:b740:a41:280::	QUIC	113	Handshake, DCID=001ac3eed4bb169670e26a7471fb0f0afde01322

Part 4: Comparing TCP and UDP

Task 1: Protocol Comparison Table

Feature	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP uses a 3-way handshake (SYN, SYN-ACK, ACK) to establish a connection before data transfer and guarantees reliable delivery using acknowledgments and retransmission mechanisms.
Data Integrity and Ordering	TCP	TCP assigns sequence numbers to packets to ensure they are reconstructed in the exact order they were sent, while also providing error-checking to guarantee data integrity.

Task 2: Use Cases and Performance Table

Criteria	TCP	UDP
Use Cases	Web browsing (HTTP/HTTPS), File transfers (FTP), Email services (SMTP/IMAP), Remote administration (SSH).	Real-time video streaming, Online multiplayer gaming, Voice over IP (VoIP), DNS queries, Live broadcasts.
Performance	Lower throughput speed and higher latency due to connection setup overhead, flow control, congestion control, and packet retransmissions.	High speed and minimal latency because it is connectionless and does not require flow control or packet delivery confirmations.